

Kubernetes for VS Code — reference analysis & design system foundations

Analysis of four infrastructure-dashboard references (Overview, Infrastructure list, Kubernetes, Deployments topology) and the foundations, components and patterns to carry into a VS Code extension for cluster management. Values are extracted from the references. v0.2 records the decisions on theme, metrics, scope, multi-cluster and stack (§12).

1. What the references get right

- **Glanceable first row.** Every screen opens with 4–6 stat tiles: uppercase mono label with a status dot, one large number, one muted qualifier ("14 · 4 acked"). Severity is carried by the number's color, not by a filled tile.
 - **Quiet surfaces, loud status.** Near-black, low-contrast layers and hairline borders; saturated color is reserved for status (green/amber/red), data series and the single primary action.
 - **Two typefaces with clear jobs.** Sans for names and values; monospace for identifiers, labels, versions, regions and metrics. Mono makes resource IDs scannable and aligns numbers.
 - **Density without clutter.** 13px rows, ~36–40px row height, inline micro-bars for CPU/memory, right-aligned numerics, relative timestamps.
 - **Visual encodings of cluster shape.** The pod-cell grid (one square per pod, colored by health) and the node pool card are the strongest Kubernetes-specific ideas — they show distribution and outliers at once.
 - **Context on demand.** Topology selection opens a floating detail card (summary, 4 metrics, dependencies, two actions) instead of navigating away.
 - **Persistent scope.** Environment (Production, amber) and provider pickers stay visible so destructive context is never ambiguous.
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2. What changes inside VS Code

The references are standalone web apps. VS Code already provides navigation, search, top chrome and a theme, so the extension must shed duplicate chrome and inherit the host.

Reference element	VS Code surface	Note
Left app sidebar	Activity Bar icon + native Tree Views	Clusters → Namespaces → Workloads. Native trees are faster and keyboard-complete.
Global search <code>⌘K</code>	Command Palette + Quick Pick	"K8s: Go to Resource...", fuzzy across kinds.
Environment / provider pickers	Status Bar item (context · namespace)	Production context tints the item with warning background.
Dashboards, tables, topology, node pools	Webview panels in the editor area	Fleet overview when several clusters are reachable; otherwise one panel per cluster. Tabs, splits and pinning come free.
Logs, events stream	Panel (bottom) — Output / Terminal / webview view	Follow mode, filter, jump-to-resource.
Resource YAML / describe	Text editor (virtual document)	Diff-before-apply uses the native diff editor.
Alerts badge	Activity Bar badge + Problems-style tree	Toasts only for SEV-1; everything else is pull.
Top bar, breadcrumbs, user avatar	Drop	Panel title + in-panel breadcrumb is enough.

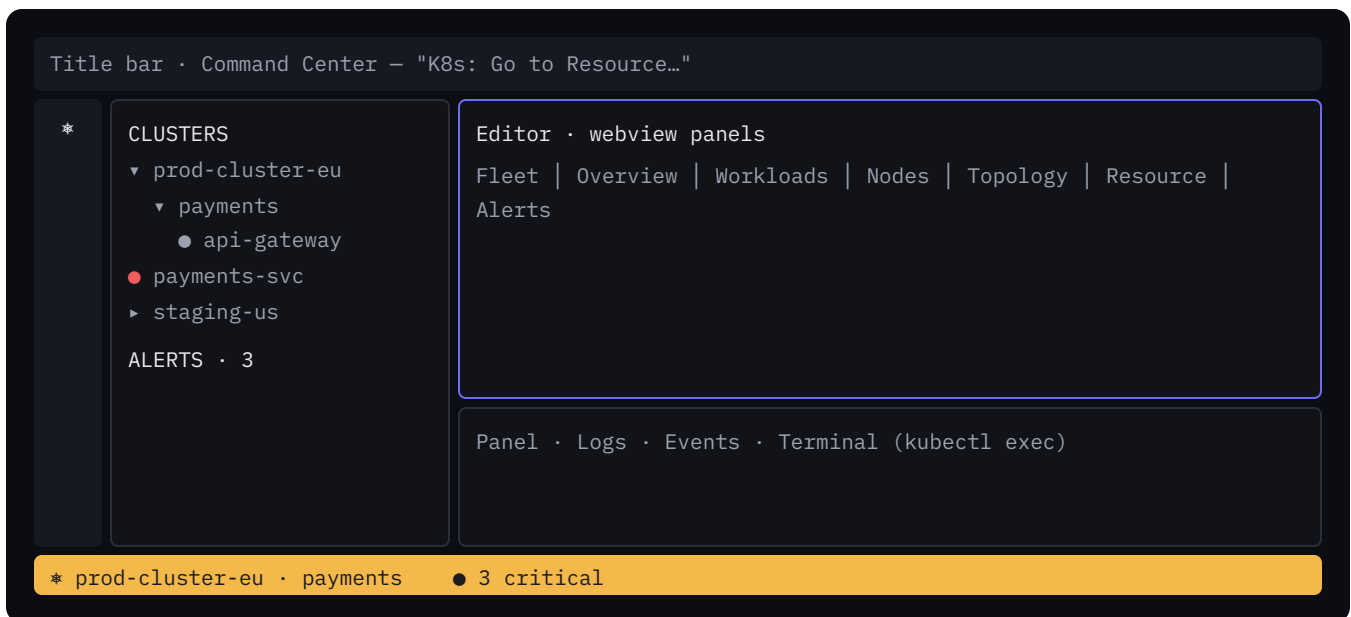


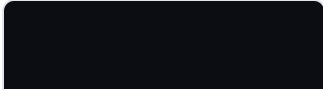
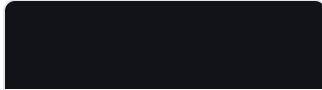
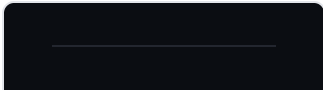
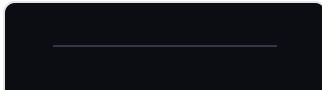
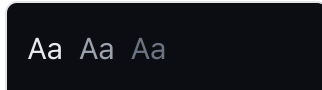
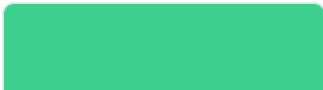
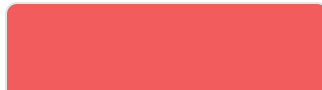
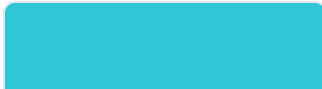
Figure 1 — Surface map. Purple outline = custom webview; everything else is native.

3. Principles

1. **Own the canvas, borrow the frame.** Webviews render in the extension's own palette — Dark by default, Light as an option — so data colors stay calibrated. Fonts, focus ring and high-contrast mode still come from VS Code so panels sit naturally in the editor.
2. **Color means state.** Green/amber/red/blue never decorate. Each status also has a glyph or word so it survives color-blind and high-contrast themes.
3. **Mono for machine strings.** Names a user would type into kubectl are mono; prose and actions are sans.
4. **Read-first, act deliberately.** Destructive actions (delete, scale to 0, restart in Production) require the context name typed or confirmed via a modal Quick Pick.
5. **Every pixel is a jump target.** Rows, pods, nodes and events open the resource; ⌘-click opens YAML.

4. Color tokens

Two extension-owned themes. Dark (default) is measured from the references; Light keeps the same roles and hue families, with status hues darkened to hold 4.5:1 on white. Selected by the `k8s.appearance` setting: `dark` | `light`. Swatches below show Dark.

 bg.canvas #0B0D12	 bg.surface #111319	 bg.raised #161921	 bg.overlay #1D2029
 border.subtle #23262F	 border.strong #343846	 text.1 / 2 / 3 #E7E9EE #9BA1AD #6B7280	 accent #6E6CF0
 status.ok #3ECF8E · Running	 status.warn #F5B84A · Degraded	 status.crit #F25C5C · Critical	 status.info #4CB8F0 · Pending
 status.off #6B7280 · Stopped	 data.1 (CPU) #2FC6D6	 data.2 (Memory) #E8C24A	 data.3 (Network) #9A87F5

Token values

Token	CSS variable	Dark (default)	Light
bg.canvas	--k8s-bg-canvas	 #0B0D12	 #F7F8FA
bg.surface	--k8s-bg-surface	 #111319	 #FFFFFF
bg.raised	--k8s-bg-raised	 #161921	 #F1F2F5
bg.overlay	--k8s-bg-overlay	 #1D2029	 #FFFFFF
border.subtle	--k8s-border	 #23262F	 #E3E5EA
border.strong	--k8s-border-strong	 #343846	 #CDD1D9
text.1	--k8s-fg	 #E7E9EE	 #14161C
text.2	--k8s-fg-muted	 #9BA1AD	 #565B66
text.3	--k8s-fg-faint	 #6B7280	 #6E7380
accent	--k8s-accent	 #6E6CF0	 #5553E0
status.ok	--k8s-ok	 #3ECF8E	 #14905E
status.warn	--k8s-warn	 #F5B84A	 #A86A12
status.crit	--k8s-crit	 #F25C5C	 #D33434
status.info	--k8s-info	 #4CB8F0	 #1F7FC2
status.off	--k8s-off	 #6B7280	 #6E7380
data.1 / 2 / 3	--k8s-data-1...3	 #2FC6D6  #E8C24A  #9A87F5	 #0E9AAB  #B88A0B  #7461E6

Borrowed from VS Code in both themes: --vscode-font-family, --vscode-editor-font-family, --vscode-font-size, --vscode-focusBorder. When VS Code runs a High Contrast theme, the extension switches to an HC token set that maps every role to --vscode-* colors and --vscode-contrastBorder, overriding k8s.appearance.

"-soft" variants (status at 14% over the surface) back status pills, environment badges and critical-row tints in both themes.

5. Typography

Webviews use the user's UI and editor fonts so the panel feels native: --vscode-font-family for sans and --vscode-editor-font-family for mono. Sizes are relative to --vscode-font-size (13px default).

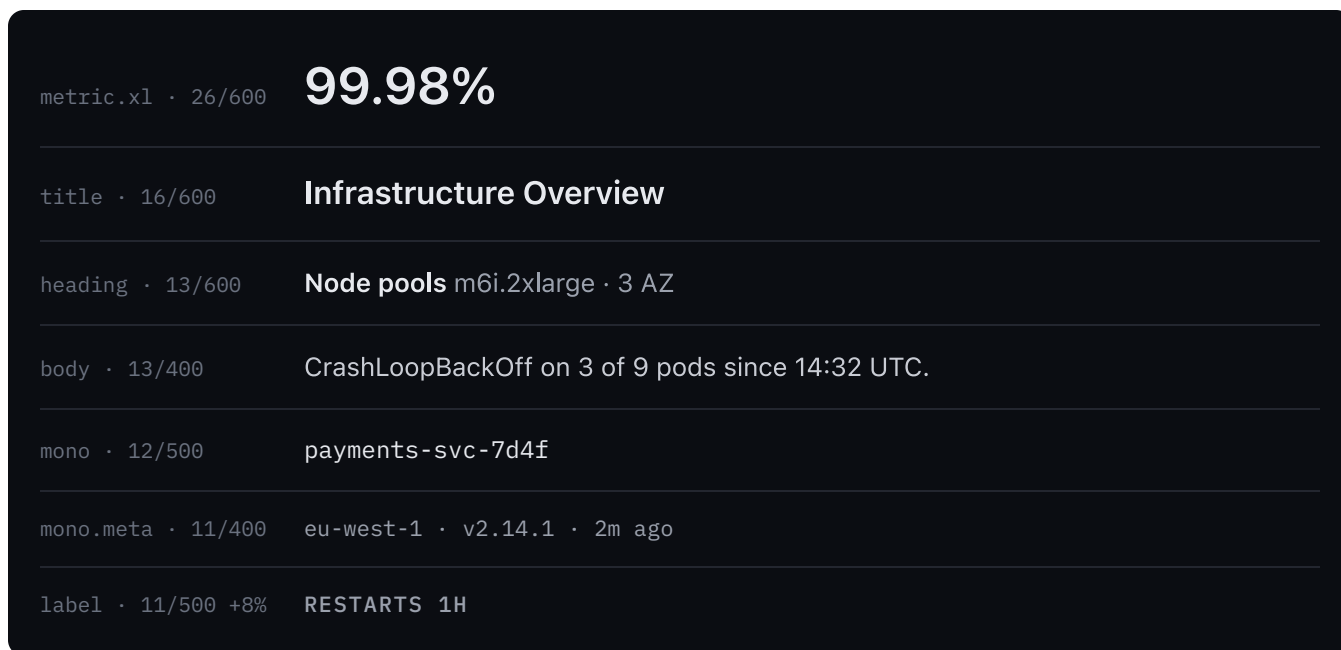


Figure 2 — Type scale. All numerics use tabular figures.

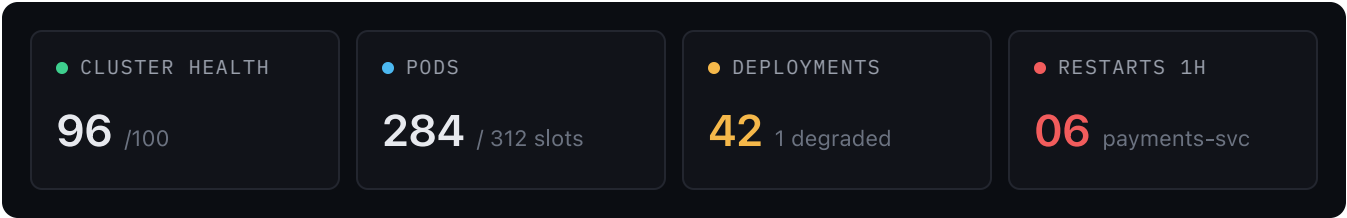
6. Space, shape, motion

- **Spacing:** 4px base — 4, 8, 12, 16, 24, 32. Tile padding 12–16; card gap 12; table cell 8×12.
- **Radius:** 4 (pills, inputs, pod cells 2), 6 (buttons, tiles), 8 (cards, popovers). Never fully round except status dots.
- **Elevation:** borders, not shadows, separate layers. Only overlays (detail card, menus) get `--vscode-widget-shadow`.
- **Density:** row 32px (compact) / 40px (comfortable), switchable in settings. Minimum webview width 360px — layouts collapse to one column below 720px.
- **Motion:** 120ms ease-out for hover/select, 200ms for overlays. Live values update in place without animation; pulse only the "LIVE" chip. Respect `prefers-reduced-motion`.

7. Core components

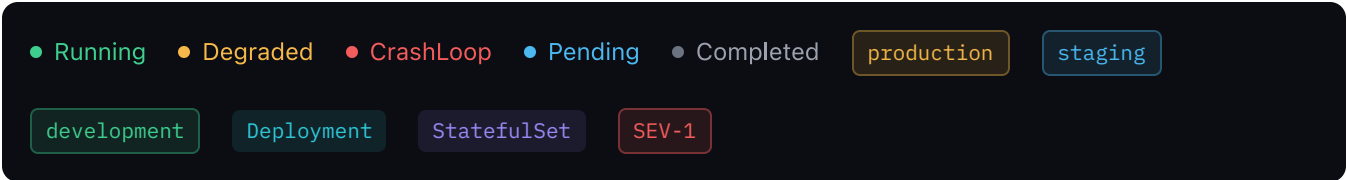
Specimens rendered with fallback values. Each maps 1:1 to a reference element.

7.1 Stat tile



Label (mono, uppercase) + dot · value colored only when non-nominal · qualifier names the culprit. Click filters the panel below.

7.2 Status, environment and kind badges



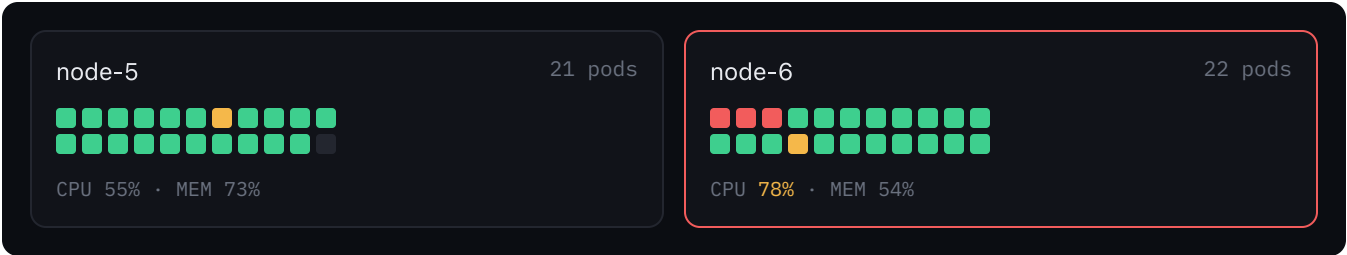
Status = dot + word (never dot alone). Pod phases map to five states; raw reason (OOMKilled, ImagePullBackOff) shows on hover and in detail. Environment badge is derived from context name rules the user configures.

7.3 Resource table row with utilization bars

NAME ↑	STATUS	NAMESPACE	CPU	MEMORY	AGE
api-gateway deploy · 12/12	● Running	edge	62%	71%	2d
payments-svc deploy · 6/9	● Critical	payments	91%	74%	38s

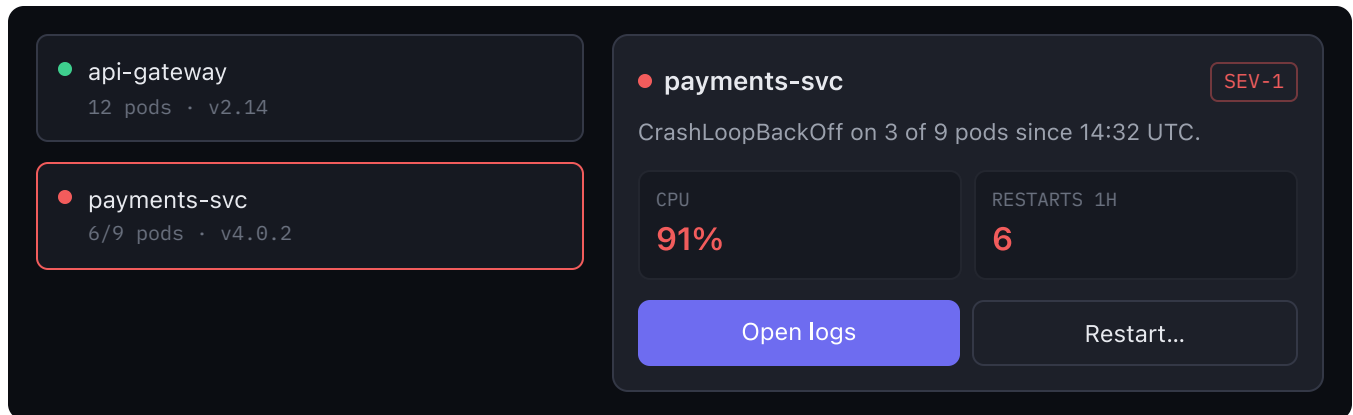
Name (sans) over kind·ready (mono). Bar turns *warn* ≥ 80% and *crit* ≥ 90% of limit; % relative to limits, not node. Critical rows get a 7% tint. Row actions (Logs, Exec, YAML, ...) appear on hover/focus only.

7.4 Node card with pod grid



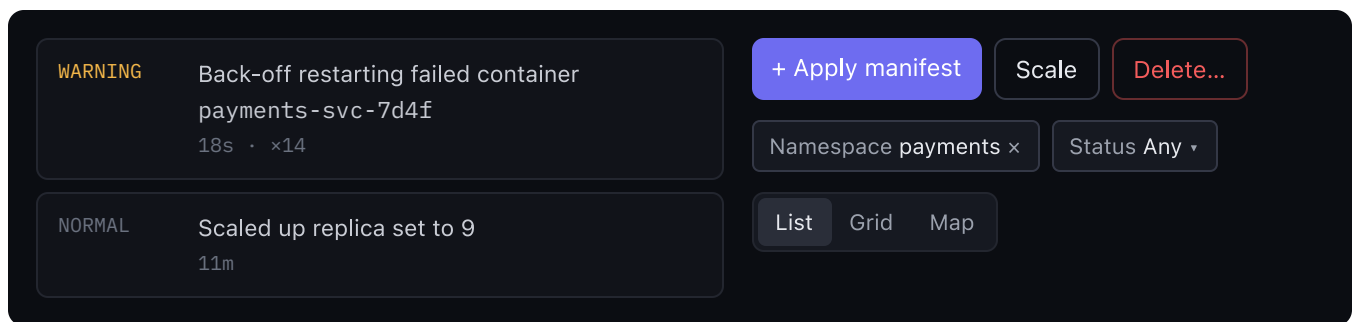
One 10px cell per pod; empty slot = border color (shows allocatable headroom). Unhealthy pods sort first. Card border takes the worst pod status. Hover a cell → pod tooltip; click → resource detail. Taints/cordon shown as a mono chip in the header.

7.5 Topology node & detail card



Topology groups by Ingress → Service → Deployment → backing store, laid out left-to-right in columns labeled in mono. Dashed edges show traffic direction with RPS labels; error-rate labels turn crit. The detail card docks top-right, never covers the selected node, and closes on Esc.

7.6 Event item, buttons, filter chips



Events dedupe by reason with a xcount. One primary button per view. Destructive buttons end in "..." and always open a confirmation. Built from Tailwind Plus button variants, recolored with the accent, border and crit tokens.

Also needed, styled from the same tokens: sparkline/area chart card (2px line in data color, 12% fill, delta chip top-right), log viewer row, empty state, skeleton loader, inline error banner, keyboard-shortcut hint (⌘K key cap).

8. Views

Each reference screen becomes an editor panel; Fleet, Resource detail and Alerts are new. Cost is out of scope.

View	Source	Composition
Fleet NEW	Overview stat row + Infrastructure list	Shown only when the kubeconfig gives access to 2+ clusters. Fleet stat row · one row per cluster (context, env badge, version, nodes, pod health bar, alerts, metrics tier) · click opens that cluster's panel.
Cluster Overview	Overview ref.	Stat row · health heatmap (one cell per workload) · CPU/Mem/Net/Error sparklines · node pools summary · recent activity. One panel per cluster; with a single cluster this is the landing view.
Workloads	Infrastructure list	Filter chips · List/Grid toggle · resource table · bulk actions bar on selection. Kind tabs replace the provider filter.
Nodes	Kubernetes ref.	Stat row · node pool grid · requests-vs-usage charts · workloads list + cluster events in a right rail (stacks below 1000px).
Topology	Deployments ref.	Canvas with columns · legend + zoom bottom · detail card on select · namespace and layer filters top-left.
Resource detail NEW	Detail card, expanded	Header (name, kind, ns, status, actions) · tabs: Summary · Pods · Events · Logs · YAML · Metrics. YAML opens a native editor with diff-on-save.
Alerts NEW	Warning/Critical tiles, events	Severity tabs with counts · grouped list (by resource) · ack / silence / open · timeline of firing state.

9. Metrics tiers

The extension detects what the cluster offers and badges the tier in the cluster header (`metrics-server` or `prometheus`). Views are designed for the lower tier first and gain detail when Prometheus is present.

Element	metrics-server	+ Prometheus
Stat tiles, util. bars, pod grid	Current CPU / memory	Same, plus trend delta
Chart cards	Session sparkline (sampled while panel is open, labeled "since opened")	Historical range picker (1h · 24h · 7d)
Request rate, error rate, p99	Hidden — single "Connect Prometheus" hint card, dismissible	Shown in Overview and detail card
Topology edges	Structure only (from Services / Ingress)	RPS and error-rate labels

Never render empty charts or zeros for missing data — use "—" with a tooltip naming the missing source.

10. Implementation — Tailwind CSS + Tailwind Plus

Webviews are built with Tailwind CSS v4. Tailwind Plus (Application UI blocks and Catalyst) is the component starting point for tables, dropdowns, dialogs, tabs, comboboxes and form controls; specialized pieces (pod grid, topology, stat tile) are custom, built from the same utilities.

- Tokens become Tailwind theme variables, so utilities read as roles: `bg-surface`, `text-fg-muted`, `border-border`, `text-crit`, `bg-ok-soft`.
- Theme switch sets `data-theme="light"` on the webview root; no `dark:` variants in markup.
- When copying Tailwind Plus code, swap its `zinc-*` / `indigo-*` classes for role tokens. Disable default palettes so stray colors fail the build.
- Interactive primitives come with Headless UI (bundled with Catalyst) — keeps keyboard and ARIA behavior in §11 for free.

```

@import "tailwindcss";

@theme {
  --color-*: initial; /* drop default palette */
  --color-canvas: var(--k8s-bg-canvas);
  --color-surface: var(--k8s-bg-surface);
  --color-fg: var(--k8s-fg);
  --color-ok: var(--k8s-ok);
  --color-ok-soft: color-mix(in srgb, var(--k8s-ok) 14%, transparent);
  --font-sans: var(--vscode-font-family);
  --font-mono: var(--vscode-editor-font-family);
  --radius-sm: 4px; --radius-md: 6px; --radius-lg: 8px;
}

:root { --k8s-bg-canvas: #0B0D12; --k8s-ok: #3ECF8E; /* ... */ }
[data-theme="light"] { --k8s-bg-canvas: #F7F8FA; --k8s-ok: #14905E; /* ... */ }

```

Code 1 — Tailwind v4 theme wired to extension tokens (abridged).

11. Accessibility & theming checks

- Verify every view in the extension's Dark and Light themes, and under VS Code High Contrast Dark and Light (HC token set: borders use `--vscode-contrastBorder`, tints removed).
- Text $\geq 4.5:1$ on its surface. *text.3* (Dark: #6B7280 on #0B0D12 $\approx 4.0:1$) is for non-essential meta only; bump to *text.2* anywhere it carries meaning.
- Full keyboard path: arrow keys in tables and pod grids, Enter opens, Space selects, Esc closes overlays; focus ring = 1px *focus* token, offset -1 .
- Pod grid and heatmap expose an accessible table alternative ("node-6: 3 critical, 1 warning, 18 running").
- Charts carry a text summary (current, delta, period) beside the plot.

12. Decisions

Question	Decision
Theme	Extension-owned palette. Dark default, Light optional (§4).
Metrics source	metrics-server baseline; Prometheus used when detected (§9).
Cost in v1	No. Cost view removed.
Multi-cluster	Fleet overview when 2+ clusters are reachable; one panel per cluster (§8).
Webview stack	Tailwind CSS v4 + Tailwind Plus (§10).